

Homework 2

Due Date: 16th of November EOD

Course: Stochastic Methods in Mathematical Modelling and AI

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Problem 1. Multivariate Gaussian Distributions (2 points)

Assuming that the joint PDF of two random variables x and y reads $P(x, y) = \frac{1}{N} \exp(-2.5x^2 - 2.5y^2 - 3xy)$ compute (a) normalisation constant N , (b) marginal PDF $P(x)$ and (c) the averages $\langle xy \rangle, \langle x^2 y^4 \rangle, \langle x^3 y^3 \rangle$ (2 points).

Problem 2. Eigenvalue distributions for Random Matrices (3 points)

Random matrix theory is applied towards a diverse set of systems ranging from physics of heavy nuclei and disordered systems to optimal control theory and deep learning. A random matrix is any matrix with random numbers as elements. In this problem you will reproduce some of the well-known classical results for Gaussian orthogonal ensemble (GOE) matrices. These matrices are symmetric and can be generated by producing the corresponding element values from $N(0, 1)$ distribution.

- **Wigner semi-circle.** Plot the spectrum of the eigenvalues of GOE matrix $N \times N$ with $N = 6000$. Check that your histogram can be well approximated by a semi-circle.
- **Wigner's surmise.** Using the same matrix you generated in the previous step check the distribution of spacing between consecutive eigenvalues. Plot the function $\frac{\pi s}{2} \exp(-\pi s^2/4)$ on top of the histogram (where s is the spacing distance)
- **Tracy-Widom.** Plot the largest eigenvalues of 1000 of 1000×1000 GOE matrices. Add the analytical curve for the Tracy-Widom distribution on top of the results.

Problem 3. KL Divergence (2 points)

Kullback–Leibler divergence D_{KL} is defined for two density functions $p(x)$ and $q(x)$ as

$$D_{KL}(p||q) = \int_{-\infty}^{\infty} p(x) \log \frac{p(x)}{q(x)} dx$$

Using Kullback–Leibler divergence check that the sum of i.i.d. uniform random variables converges to a normal distribution $\mathcal{N}(0, 1)$. At what point, in terms of the number of summands N and D_{KL} , does this sum become indistinguishable from sample from a Gaussian distribution?

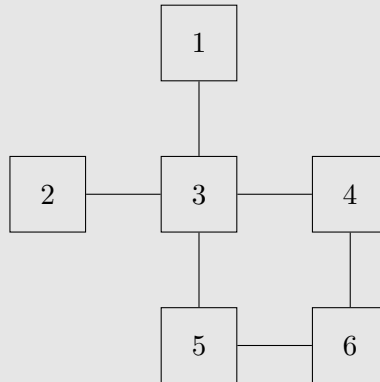
1. Write a function that produces a sum $Z = X_1 + \dots + X_N$, where $X_i \in \text{Uniform}[0, 1]$. Normalize the sum according to the statement of the Central Limit Theorem.
2. Plot a PDF of Z and ensure it becomes closer to normal when N increases.
3. Compute the $D_{KL}(P||Q)$ where P is the PDF of Z and Q is the density function of the normal distribution for each number of summands N .
4. Plot $D_{KL}(P||Q)$ values for $N = 1, \dots, 25$.

Hint: the PDF of the sum could be obtained through the convolution, or, easier, as values of the histogram if a sample for X_i is big enough.

Problem 4 Markov's nightmare (3 points)

Barry is a chronically sleep-deprived man who, every night at 3:17 AM, begins sleepwalking through his bizarrely arranged six-room apartment. He has no memory of where he's going, but his feet move with eerie precision... and zero regard for logic or safety.

Each room has doors leading to adjacent rooms (see diagram below). At each step, Barry randomly picks *one* door to walk through. He never stops. Ever. (Insomnia is relentless.)



1. Give the transition matrix P for this Markov chain.
2. Show that it is irreducible but not aperiodic.
3. Find the stationary distribution.
4. Now suppose that a cup of freshly brewed coffee is placed in kitchen (Room 5). Barry starts in his bedroom (Room 1). Find the expected number of steps before Barry reaches the kitchen for the first time.
5. After sipping the coffee, Barry awakens. Yet, a fleeting moment later, he resumes his sleepwalking as if nothing happened. Find the expected time for Barry to return to his bedroom (Room 1).